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Accelerating Molecular Generation through Representation Alignment

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Declaration

I, Shreyas Ravishankar of Hughes Hall, being a candidate for the Master of Philosophy in Advanced Computer Science, hereby declare that this report and the work described in it are my own work, unaided except as may be specified below, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose. In preparation of this report, I adhered to the Department of Computer Science and Technology AI Policy. [The project required the approved use of {insert name of technology} and such use is acknowledged in the text at the relevant sections.] [I am content for my report to be made available to the students and staff of the University.]

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Abstract

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Acknowledgements

This project would not have been possible without the wonderful support of ... [optional]

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Chapter 1

Introduction

The story of generative machine learning in scientific domains has been a dance between data scale, data quality, and structural priors. Our thesis studies a recent addition to this story, alignment to representations coming from an external encoder, and investigates its effectiveness when applied to molecular generation. We study this technique in two regimes, small molecules and protein backbones, and characterise its utility along two axes. First, does the encoder carry transferable structural information that the generator does not learn from the denoising objective? Second, does the evaluation surface have headroom for any gain to register?

In this introduction, we trace the narrative arc of molecular generation to the point that motivates our thesis: a need for data-efficient modelling techniques that preserve generation quality given the shift towards prior-light model architectures.

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow-matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de novo design challenges across scientific fields, such as generating protein backbones [15, 16, 11], inorganic materials [18], and drug-like small molecules [12]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of training examples [24]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [11], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [30]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [31]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

However, a recent wave of work has moved entirely the other way: drop hard priors, use bare-bones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine learning, across domains, on leveraging representation instead of model architecture to inject soft inductive biases for structured data. Vision Transformers [32] showed that the spatial-locality and translation-equivariance priors of CNNs can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [33] have made the same case for graphs. AlphaFold 3 [35] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [34], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [36] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. Tabasco [12] and Proteina [11], which are vanilla transformer-based generators for small-molecules and protein backbones respectively, are explicitly designed to facilitate data and model scale.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [11, 26] to expand training corpora into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [28], and the theoretical sequence space at a median protein length [29] is hundreds of orders of magnitude greater still. Even at the frontier of the field’s data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity and curated data for synthetic scale. And yet, this may not be enough. This motivates the need for new techniques that make training still more efficient without sacrificing quality.

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [3], who argue that these models suffer from a *representation bottleneck*. During training, the model is implicitly asked to learn two distinct tasks simultaneously: (1) representation, learning a semantically meaningful representation of complex data; and (2) generation, learning how to construct data from that representation. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets we see in practice.

Yu et al. propose Representation Alignment (REPA) as a remedy. The technique adds a regularisation term that aligns a generator’s intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a 17.5× training speed-up against a strong baseline. REPA has since been extended beyond images to video [7], audio [8], and recently to molecular generation [5, 6]. However, whether REPA’s gains transfer to molecular settings, and why, remain uncharacterised.

In this vein, the goal of our report is to contribute methodological insights about represen-

tation learning in molecular generative models. The core question we investigate is: *can these models benefit, in terms of training efficiency and generation quality, from representation alignment with pre-trained models?* We aim to characterise when and why this works or does not, including failure modes and limitations. We note that this question is especially interesting

The report contributes three pieces of work.

- (i) An encoder-profiling framework characterising candidate encoders along three axes — information content, reachability, and trainability — applied to seven candidate encoders across protein and small-molecule domains. The framework operationalises encoder quality in domain-agnostic statistical terms (effective rank, linear-probe recoverability), complementing but not duplicating the vision-specific structural metrics proposed by Singh et al. [4].
- (ii) Paired empirical studies of REPA on two flow-based molecular generators: Tabasco [12] for small molecules, and Proteina [11] for unconditional protein backbone generation. The studies include ablations over alignment-layer choice, projector depth, averaging mode, loss weight, and combination mode, alongside convergence sweeps and seed-variance analysis.
- (iii) A characterisation of what governs REPA’s effectiveness in molecular generation, framed through the three under-explored axes identified above — encoder choice, evaluation-surface headroom, and a domain-general analogue of the structural encoder-quality argument. We offer this characterisation as a working framework with encoder-selection heuristics for future domains.

We do not propose a new model architecture, alignment loss, or evaluation metric. We do not establish the conditions under which REPA helps as universally necessary or sufficient: our two studies amount to one regime in which the conditions are weak and REPA does not visibly help, and one in which they are met alongside other favourable factors and REPA does help. That is consistent with the framework, but it is not a controlled experiment over the conditions themselves.

The remainder of the report is organised as follows. Chapter 2 sets out the background needed to read the rest: flow matching, REPA, and the parameterisations we use for small molecules and proteins. Chapter 3 places the work in context, covering the scaling-over-priors trend in structural biology and the recent literature on representation alignment in and out of vision. Chapter 4 discusses what it means for a generation to be “good” in each domain, and the tensions that question hides. Chapter 5 develops the encoder-profiling framework and uses it to predict where REPA should and should not help. Chapters 6 and 7 present the two paired studies, small molecules first and protein backbones second. Chapter 8 brings the evidence together.

Chapter 2

Background

Chapter 3

Related work

Chapter 4

Evaluating molecular generation

Chapter 5

Encoder targets and profiling

Chapter 6

REPA on small molecules

Chapter 7

REPA on protein backbones

Chapter 8

Conclusions

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Appendix A

Technical details

A.1 Placeholder section

TODO.